

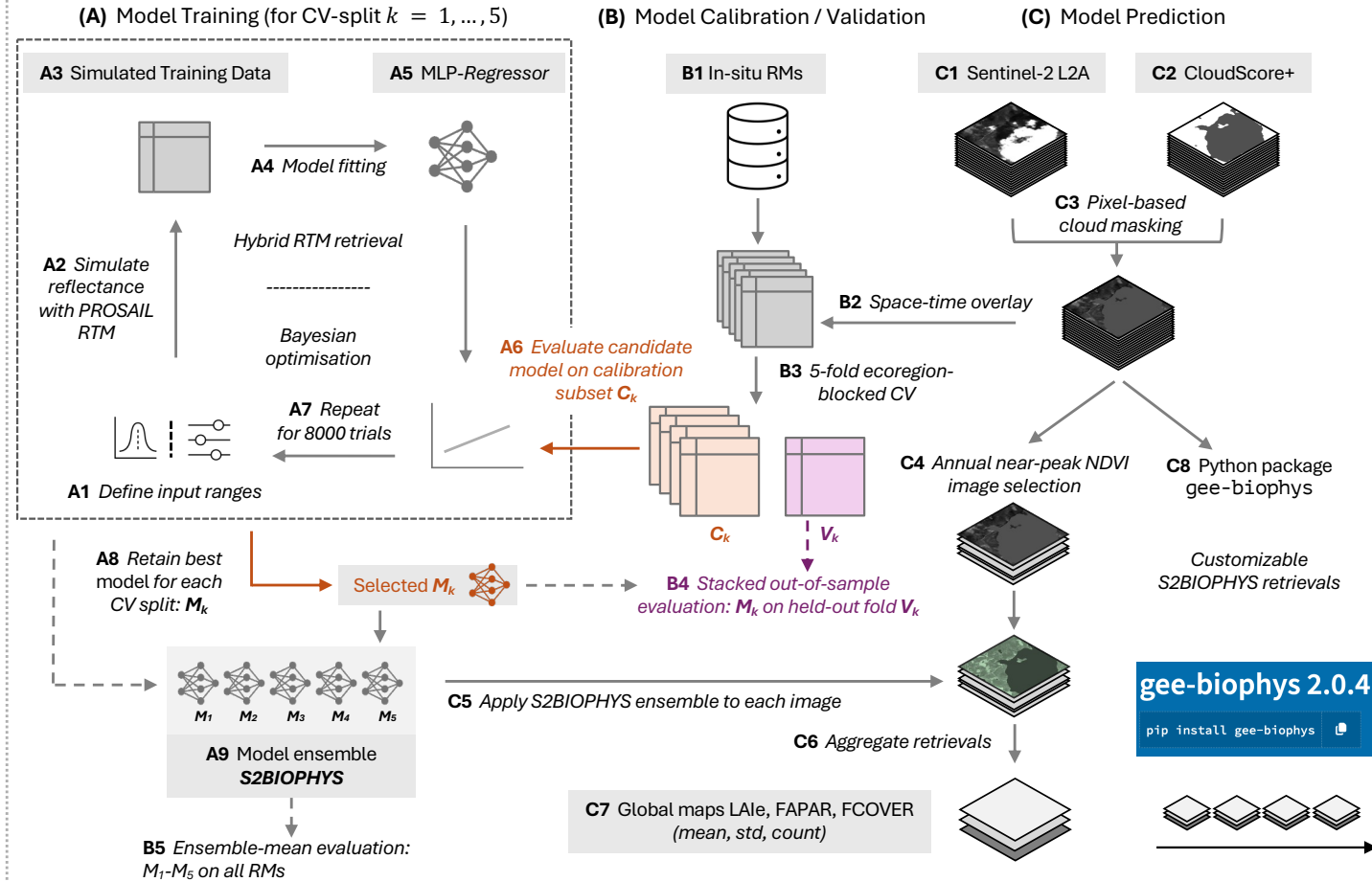


Figure 1: Overview of Model Training and Prediction Workflow

A The core of the model training workflow revolves around the iterative optimisation of the parameters involved in a PROSAIL RTM hybrid inversion procedure. First, we defined input trait ranges and distributions (**A1**) and then simulated reflectances using PROSAIL for each trait combination (**A2**). The simulated reflectances (**A3**) served as training data for fitting (**A4**) a multi-layer perceptron regression model (**A5**). We evaluated the model fit using the calibration split of the in-situ data (**A6**) and repeated this procedure for 1000 iterations. We used the Optuna hyperparameter optimisation framework with the search space defined in Table 2.

B We obtained in-situ reference measurements from GBOV (**B1**) for LAI, FAPAR, and FCOVER and overlaid them with the closest available cloud-free Sentinel-2 reflectances (**B2**). For each biophysical property, we performed a 5-fold stratified CV split based on ecoregions, ensuring observations from the same ecoregion appeared only in either calibration or validation splits (**B3**). For each CV split, we repeated the optimisation of the PROSAIL RTM hybrid inversion procedure and stacked (**A8**) the best models into a model ensemble (**A9**). We evaluated the model performance on held-out ecoregions using the validation split (**B4**).

C For global predictions, we used the Sentinel-2 L2A data catalog (**C1**) and joined it with the precomputed CloudScore+ collection (**C2**) to mask clouds at the pixel level (**C3**). To represent the vegetative period throughout the year, we calculated mean scene-level NDVI and retained scenes that balanced lower cloud coverage and higher NDVI (**C4**). We randomly assigned each Sentinel-2 scene to a model ensemble member (**A9**), predicted the scene (**C5**) and aggregated the predictions to compute mean, standard deviation, and acquisition count at the pixel level (**C6**). The resulting global biophysical maps (**C8**) are available for LAI, FAPAR, and FCOVER at 1000m, 100m, and 20m resolutions from 2019 to 2024.